

Unit 4. Dynamics

4.1 The Concept of Force and Newton's Laws of motion

At the end of completion of this section you should be able to:

- discuss what the concept of force means in physics, and why forces are vectors.
- define force and discuss its effects on a body.
- distinguish between contact forces and field forces.
- discuss the four fundamental forces in nature and their real life application.
- state newton's laws of motion
- demonstrate the cause effect relationship of force and acceleration.
- apply newton's second law to determine the weight of an object.
- demonstrate the relationship between weight and normal force.

4.2 Frictional Force

At the end of completion of this section you should be able to:

- describe the general characteristics of friction
- list the various types of friction
- calculate the magnitude of static and kinetic friction, and use these in
- problems involving Newton's laws of motion
- solve more complex acceleration problems involving Newton's laws of motion.
- use free-body diagrams to solve problems on Newton's Laws of motion.

4.3 The First Condition of Equilibrium

At the end of completion of this section you should be able to:

- define the term equilibrium
- explain the first condition of equilibrium
- apply the first condition of equilibrium to solve practical problems.

4.4 Work, Energy and Power

By the end of this section, you will be able to:

- define work in terms of the applied force and the displacement.
- calculate the work done by a constant force, by a varying force.
- demonstrate by examples that maximum work is done when the applied force is parallel to the displacement vector.
- apply work-energy theorem to solve problems.
- calculate kinetic energy and potential energy of a body.
- differentiate between conservative and non-conservative forces.
- explain the law of the conservation of energy.
- solve problems by applying the law of conservation of energy.
- define power and solve problems involving power.

4.5 Conservation of mechanical energy

By the end of this section, you will be able to:

- define mechanical energy.
- state the law of conservation of mechanical energy.
- apply the law of conservation of mechanical energy to solve problems

4.6 Impulse and Linear Momentum

By the end of this section, you will be able to:

- Discuss impulse and linear momentum
- relate impulses to collisions
- apply the impulse-momentum theorem to solve problems

- differentiate elastic and inelastic collisions
- state the law of conservation of linear momentum
- explain the meaning and usefulness of the concept of center of mass
- calculate the center of mass of a given system
- calculate the velocity of the center of mass
- apply the law of conservation of linear momentum to solve problems involving collision in one dimension.